



Digital Twin Prototype for Consumer Value Creation in a Smart Living Environment

Project Overview

This proof-of-concept prototype investigates how a digital twin of a physical living environment can support consumer value creation, personalization, and data-driven service decision-making. It demonstrates how environmental and consumer data can be integrated to generate actionable insights for business renewal and service innovation.

Digital twins are increasingly recognized as virtual representations of real-world environments that support monitoring, simulation, and informed decision-making. In this project, a living room was reconstructed using Polycam as an equivalent tool for photorealistic 3D reconstruction to create a navigable digital twin. The objective was not only to develop a photorealistic virtual model but also to demonstrate how a consumer environment can become a data-enabled space for generating actionable insights and supporting value creation.

Prototype Development

The prototype includes a navigable 3D reconstruction and a simulated data layer embedded into the scene. The overlays include temperature, occupancy level, smart device activity, consumer engagement score, content consumption duration, comfort score, satisfaction index, and personalization readiness. Since this is an early-stage proof of concept rather than a live IoT deployment, the data is realistically simulated with small variations to represent dynamic real-world conditions.

Decision Layer and Business Value

A rule-based decision layer was added to convert the data overlays into actionable recommendations. For example, high engagement, active smart TV usage, and longer content consumption suggest an opportunity to personalize entertainment recommendations. Similarly, strong comfort and satisfaction scores suggest maintaining the current environment while introducing targeted service enhancements. These recommendations illustrate how digital twins can transform observational data into actionable managerial insights that enhance consumer experiences, support business renewal, and improve value creation.

Application to B2B and Service Contexts

In a B2B or service context, this prototype could be extended to hotels, smart offices, retail showrooms, or university learning spaces. For example, a hotel could use room-level digital twins to understand guest comfort, service engagement, and personalization readiness. A retail or service firm could use similar data to adapt layouts, personalize offers, improve customer journeys, and support business model renewal. This aligns with service-dominant logic, where value is co-created through interaction between users, firms, technologies, and service environments.

Future Research Directions

The prototype also highlights ethical and methodological considerations. Future versions should integrate live IoT sensors, validated consumer data, privacy safeguards, and machine learning models. Future research could empirically evaluate how digital twins influence consumer engagement, value co-creation, adaptive business models, and sustainable business renewal across service industries. This proof-of-concept prototype provides a foundation for future empirical research on digital twins, consumer value creation, and service innovation.

References

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